Economics of Organizations and Incentives

MPE – 4th term, 2026

Lecture 1: Static games of complete information I



Isabela Brandão Furtado
isabelabf@insper.edu.br

What is game theory?

Game theory studies **mathematical models** of how **rational** agents make decisions when the **outcome** of each agent's choice **depends on the actions of others**.

What is game theory?

Game theory studies **mathematical models** of how **rational** agents make decisions when the **outcome** of each agent's choice **depends on the actions of others**.

- **Models:** they abstract from real behavior to reveal the logic behind strategic decisions.

What is game theory?

Game theory studies **mathematical models** of how **rational** agents make decisions when the **outcome** of each agent's choice **depends on the actions of others**.

- **Models:** they abstract from real behavior to reveal the logic behind strategic decisions.
- **Rationality:** as in Microeconomics I, choices follow from players' preferences, represented by an expected-utility function (methodological individualism).

What is game theory?

Game theory studies **mathematical models** of how **rational** agents make decisions when the **outcome** of each agent's choice **depends on the actions of others**.

- **Models:** they abstract from real behavior to reveal the logic behind strategic decisions.
- **Rationality:** as in Microeconomics I, choices follow from players' preferences, represented by an expected-utility function (methodological individualism).
- **Interdependence:** unlike in Microeconomics I, each player's payoff depends on the actions of *all* players.

What is game theory?

Each player must form expectations about the **actions of the other players**.

- But their actions depend on **their expectations about my actions**.

What is game theory?

Each player must form expectations about the **actions of the other players**.

- But their actions depend on **their expectations about my actions**.
- Reasoning about reasoning: **strategic reasoning**.

What is game theory?

Each player must form expectations about the **actions of the other players**.

- But their actions depend on **their expectations about my actions**.
- Reasoning about reasoning: **strategic reasoning**.
- Players form beliefs about each other's choices, and strategic reasoning requires understanding these mutual expectations.

What is game theory?

Each player must form expectations about the **actions of the other players**.

- But their actions depend on **their expectations about my actions**.
- Reasoning about reasoning: **strategic reasoning**.
- Players form beliefs about each other's choices, and strategic reasoning requires understanding these mutual expectations.
- Game theory formalizes the interaction between preferences and expectations.

Examples

- **Partnership:** two partners simultaneously choose whether to exert effort E, at cost $c > 0$, or to shirk S. Output equals y times the number of partners who exert effort and is split equally.
- **Cournot duopoly:** two firms produce a homogeneous good and face a downward-sloping demand curve. Each firm simultaneously chooses its output $q_i \geq 0$.
- **Second-price auction:** $n \geq 2$ buyers compete for a single indivisible object. Bids are submitted simultaneously; the highest bidder wins and pays the *second-highest* bid.

About this course

This course: topics

Lecture	Date	Topic
1	Oct 13	Static games: dominance, iterated elimination
2	Oct 20	Static games: Nash equilibrium, mixed strategies
3	Oct 27	Dynamic games: backward induction, subgame perfection
4	Nov 3	Multistage games and bargaining
5	Nov 6 (Fri)	Repeated games I: cooperation
6	Nov 10	Repeated games II: collusion, relational contracts
7	Nov 17	Bayesian games, adverse selection and auctions
8	Nov 24	Signaling and perfect Bayesian equilibrium
9	Dec 1	Short-paper presentations
10	Dec 8	Final exam

This course: learning goals

By the end of the course, you should be able to:

- recognize the main classes of games and the central results of game theory, and understand why they matter;
- solve games of every major type, choosing the appropriate tools according to *when* each agent moves and *what* each agent knows;
- model real-world strategic interactions as games, making the assumptions explicit and drawing insights about the situation at hand.

This course: assessment

Component	Weight
In-class exercises	10%
Short paper (groups)	35%
Content	15%
Presentation and Q&A (Lecture 9)	15%
Comments on another group's paper	5%
Final exam (individual, Lecture 10)	55%

The short paper includes a motivation, a brief literature review, a research question and an economic model. *Note:* the schedule for partial submissions will be agreed with the class.

Representing a game

Example 1: let's play a game!

The grade game. Without showing your neighbor, write down either the letter α or the letter β on a sheet of paper. Think of it as a "grade bid". Your sheet will be randomly paired with another one, and neither of you will ever know who your pair was. Grades are assigned as follows:

- if you write α and your pair writes β, you get an A and your pair gets a C;
- if both of you write α, you both get a B$-$;
- if you write β and your pair writes α, you get a C and your pair gets an A;
- if both of you write β, you both get a B$+$.

Representation: the outcome matrix

A bimatrix:

- rows are the strategies of player 1 (you);
- columns are the strategies of player 2 (your pair);
- each cell shows the grades of (player 1, player 2).

Representation: the outcome matrix

A bimatrix:

- rows are the strategies of player 1 (you);
- columns are the strategies of player 2 (your pair);
- each cell shows the grades of (player 1, player 2).

		Pair: α	Pair: β
You	α	B−, B−	A, C
	β	C, A	B+, B+

Representation: the outcome matrix

A bimatrix:

- rows are the strategies of player 1 (you);
- columns are the strategies of player 2 (your pair);
- each cell shows the grades of (player 1, player 2).

		Pair: α	Pair: β
You	α	B−, B−	A, C
	β	C, A	B+, B+

To model this situation as a game, grades must be translated into payoffs (utility levels). **How does each person value their own grade, and the grade of others?**

Representation: the outcome matrix

One possible set of payoffs:

		Pair: α	Pair: β
You	α	$0,\ 0$	$3,\ -1$
	β	$-1,\ 3$	$1,\ 1$

Normal-form (strategic-form) games

A normal-form game is a triple

$$G = (I,\ \{S_i\}_{i \in I},\ \{u_i\}_{i \in I}),$$

where

- $I = \{1, 2, \ldots, n\}$ is the set of players;

Normal-form (strategic-form) games

A normal-form game is a triple

$$G = (I,\ \{S_i\}_{i \in I},\ \{u_i\}_{i \in I}),$$

where

- $I = \{1, 2, \ldots, n\}$ is the set of players;
- S_i is the set of (pure) strategies of player i;

Insper

Normal-form (strategic-form) games

A normal-form game is a triple

$$G = (I,\ \{S_i\}_{i \in I},\ \{u_i\}_{i \in I}),$$

where

- $I = \{1, 2, \ldots, n\}$ is the set of players;
- S_i is the set of (pure) strategies of player i;
- $u_i : S_1 \times \cdots \times S_n \to \mathbb{R}$ is player i's payoff function.

Notation and terminology

- $S = S_1 \times \cdots \times S_n$ is the Cartesian product of the strategy sets.
- A strategy profile is $s = (s_1, \ldots, s_n) \in S$.
- For $i \in I$, $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$ denotes the strategies of everyone except i, and S_{-i} the set of such profiles.
- For $s_i \in S_i$ and $s_{-i} \in S_{-i}$, we write (s_i, s_{-i}) for the full profile.

Static vs. dynamic choices

- In normal-form games, choices are *static*: each player chooses a strategy without observing the others' choices.
- Many strategic situations, however, are inherently dynamic.

Static vs. dynamic choices

- In normal-form games, choices are *static*: each player chooses a strategy without observing the others' choices.
- Many strategic situations, however, are inherently dynamic.
- **Key fact:** any situation, static or dynamic, can be represented as a normal-form game by suitably defining strategies. This will become clear when we study the extensive form.

Complete vs. incomplete information

In a game of **complete information**, the game being played is *common knowledge*. That is, all players know:

- who is playing;
- which actions are available to each player;
- how outcomes translate into payoffs.

Definition

An event E is **common knowledge** if (1) everyone knows E, (2) everyone knows that everyone knows E, and so on, *ad infinitum*.

What is a strategy?

- A strategy is a **complete contingent plan of action**: it prescribes what to do in every contingency that may arise.

What is a strategy?

- A strategy is a **complete contingent plan of action**: it prescribes what to do in every contingency that may arise.
- Example: a chess player who decides in advance which move to make in every possible board position.

What is a strategy?

- A strategy is a **complete contingent plan of action**: it prescribes what to do in every contingency that may arise.
- Example: a chess player who decides in advance which move to make in every possible board position.
- In this sense, even long dynamic interactions can be encoded as a one-shot choice of strategies.

What is a payoff?

- $u_i : S \to \mathbb{R}$ is player i's utility over outcomes (strategy profiles).
- u_i represents player i's preferences over outcomes.
- In some lectures we will impose stronger conditions (the expected-utility axioms).

Example 2: writing the grade game in normal form

Write the grade game in normal form, $G = (I, \{S_i\}_{i \in I}, \{u_i\}_{i \in I})$, using the payoffs below.

		Pair: α	Pair: β
You	α	$0,\ 0$	$3,\ -1$
	β	$-1,\ 3$	$1,\ 1$

Dominance

What are we looking for?

- A **solution concept** is a method for analyzing games that singles out their most reasonable outcomes.

What are we looking for?

- A **solution concept** is a method for analyzing games that singles out their most reasonable outcomes.
- An **equilibrium** is a strategy profile selected by a solution concept; it is the theory's prediction.

What are we looking for?

- A **solution concept** is a method for analyzing games that singles out their most reasonable outcomes.
- An **equilibrium** is a strategy profile selected by a solution concept; it is the theory's prediction.
- To be useful, a theory must apply to a wide range of strategic situations.

What are we looking for?

Three criteria to evaluate solution concepts:

- **Existence: how often does it apply?** A good solution concept should yield at least one equilibrium in a wide class of games.

What are we looking for?

Three criteria to evaluate solution concepts:

- **Existence: how often does it apply?** A good solution concept should yield at least one equilibrium in a wide class of games.
- **Uniqueness: how much does it restrict behavior?** It should narrow outcomes down to a small, reasonable set. Uniqueness is hard to achieve: since each player's best action depends on the others', games often have multiple equilibria.

What are we looking for?

Three criteria to evaluate solution concepts:

- **Existence: how often does it apply?** A good solution concept should yield at least one equilibrium in a wide class of games.
- **Uniqueness: how much does it restrict behavior?** It should narrow outcomes down to a small, reasonable set. Uniqueness is hard to achieve: since each player's best action depends on the others', games often have multiple equilibria.
- **Invariance: how sensitive is it to small changes?** A robust solution concept should not change its prediction when payoffs change by a small amount ε.

Dominance

Let $s_i, s_i' \in S_i$ be strategies of player i.

- s_i is **strictly dominated** by s_i' if, for all $s_{-i} \in S_{-i}$,

$$u_i(s_i, s_{-i}) < u_i(s_i', s_{-i}).$$

- s_i is **weakly dominated** by s_i' if, for all $s_{-i} \in S_{-i}$,

$$u_i(s_i, s_{-i}) \leq u_i(s_i', s_{-i}),$$

  with strict inequality for at least one s_{-i}.
- A strategy is strictly (weakly) **dominant** if it strictly (weakly) dominates every other strategy of that player.

Example: the prisoner's dilemma

The police can only prove a minor crime unless one of two suspects confesses. Each suspect is questioned separately and offered a deal: confess (*fink*, F) in exchange for a reduced sentence, or stay silent (*mum*, M). The outcome depends on both suspects' choices.

		Player 2	
		M	F
Player 1	M	$-2,\ -2$	$-5,\ -1$
	F	$-1,\ -5$	$-4,\ -4$

Dominant-strategy equilibrium

Definition

The strategy profile $s^D \in S$ is a **strict dominant-strategy equilibrium** if, for every $i \in I$, s_i^D is a strictly dominant strategy.

Dominant-strategy equilibrium

Definition

The strategy profile $s^D \in S$ is a **strict dominant-strategy equilibrium** if, for every $i \in I$, s_i^D is a strictly dominant strategy.

Proposition

If the game G has a strict dominant-strategy equilibrium s^D, then s^D is its *unique* dominant-strategy equilibrium.

Dominant-strategy equilibrium

Definition

The strategy profile $s^D \in S$ is a **strict dominant-strategy equilibrium** if, for every $i \in I$, s_i^D is a strictly dominant strategy.

Proposition

If the game G has a strict dominant-strategy equilibrium s^D, then s^D is its *unique* dominant-strategy equilibrium.

Lesson 1: a rational player never plays a strictly dominated strategy.

Dominance and social dilemmas

Definition (Pareto optimality)

A strategy profile $s \in S$ **Pareto dominates** $s' \in S$ if $u_i(s) \geq u_i(s')$ for all $i \in I$ and $u_i(s) > u_i(s')$ for at least one $i \in I$. A strategy profile is **Pareto optimal** if no other strategy profile Pareto dominates it.

Dominance and social dilemmas

Definition (Pareto optimality)

A strategy profile $s \in S$ **Pareto dominates** $s' \in S$ if $u_i(s) \geq u_i(s')$ for all $i \in I$ and $u_i(s) > u_i(s')$ for at least one $i \in I$. A strategy profile is **Pareto optimal** if no other strategy profile Pareto dominates it.

Lesson 2: individually rational choices can lead to socially undesirable outcomes.

Example 3: dominant and dominated strategies

Does any player have a **dominant** strategy? Are there **dominated** strategies?

		Player 2		
		L	C	R
Player 1	U	$4, 3$	$5, 1$	$6, 2$
	M	$2, 1$	$8, 4$	$3, 6$
	D	$3, 0$	$9, 6$	$2, 8$

Iterated elimination of dominated strategies

- We have seen that a rational player never plays a strictly *dominated* strategy.

Iterated elimination of dominated strategies

- We have seen that a rational player never plays a strictly *dominated* strategy.
- If all players know this, each of them can ignore the strictly dominated strategies of the others...

Iterated elimination of dominated strategies

- We have seen that a rational player never plays a strictly *dominated* strategy.
- If all players know this, each of them can ignore the strictly dominated strategies of the others...
- ...and the others can reason in the same way.

Iterated elimination of dominated strategies

- We have seen that a rational player never plays a strictly *dominated* strategy.
- If all players know this, each of them can ignore the strictly dominated strategies of the others…
- …and the others can reason in the same way.
- This works because rationality is *common knowledge*.
- Dominated strategies can therefore be eliminated iteratively.

Iterated elimination of dominated strategies

Let $S_i^0 = S_i$. For each $k \geq 1$ and each player i, define recursively

$$S_i^k = \{s_i \in S_i^{k-1} : \nexists s_i' \in S_i^{k-1} \text{ such that } u_i(s_i, s_{-i}) < u_i(s_i', s_{-i}) \ \ \forall s_{-i} \in S_{-i}^{k-1}\}.$$

The set of strategies that survive every round is $S_i^\infty = \bigcap_{k \geq 1} S_i^k$.

Iterated elimination of dominated strategies

Let $S_i^0 = S_i$. For each $k \geq 1$ and each player i, define recursively

$$S_i^k = \{s_i \in S_i^{k-1} : \nexists s_i' \in S_i^{k-1} \text{ such that } u_i(s_i, s_{-i}) < u_i(s_i', s_{-i}) \ \ \forall s_{-i} \in S_{-i}^{k-1}\}.$$

The set of strategies that survive every round is $S_i^\infty = \bigcap_{k \geq 1} S_i^k$.

Definition

Any strategy profile $s^{ES} = (s_1^{ES}, \ldots, s_n^{ES})$ that survives IESDS is an **iterated-elimination equilibrium**.

Iterated elimination of dominated strategies

Let $S_i^0 = S_i$. For each $k \geq 1$ and each player i, define recursively

$$S_i^k = \{s_i \in S_i^{k-1} : \nexists s_i' \in S_i^{k-1} \text{ such that } u_i(s_i, s_{-i}) < u_i(s_i', s_{-i}) \ \ \forall s_{-i} \in S_{-i}^{k-1}\}.$$

The set of strategies that survive every round is $S_i^\infty = \bigcap_{k \geq 1} S_i^k$.

Definition

Any strategy profile $s^{ES} = (s_1^{ES}, \ldots, s_n^{ES})$ that survives IESDS is an **iterated-elimination equilibrium**.

Proposition

If s^* is a strict dominant-strategy equilibrium, then s^* is the unique profile that survives IESDS.

Application: spatial competition

Motivation: a pharmacy rivalry in Brasília



Why do pharmacies of competing chains open right next to each other?

Source: Revista PEGN (2023): rixa de farmácias em rua de Brasília viraliza nas redes

Mini case: where to open the pharmacy?

Two pharmacy chains, A and B, will each open one store on a new avenue with 7 blocks.



- Each block has the same number of customers (normalize to 1 per block).
- Both chains choose a block **simultaneously**. Prices and products are identical.
- Customers buy from the **nearest** store and split evenly if both are equally distant.
- Each chain maximizes its number of customers.

Mini case: questions (in pairs, hand in today)

(a) Write the game in normal form. Give a formula for $u_A(s_A, s_B)$ when $s_A < s_B$, $s_A = s_B$ and $s_A > s_B$.

(b) Show that block 1 is strictly dominated by block 2. Is any block strictly dominant?

(c) Apply IESDS. Describe each round. Which location profile survives?

(d) Customers dislike walking. Which pair of locations minimizes the total distance walked? Compare with your answer in (c).

(e) Which assumptions drive your prediction? What would change if the chains also chose prices? Give a real-world example of clustering, or of firms spreading out.

Wrap-up and next lecture

Today

- Normal-form games; static vs. dynamic games; complete vs. incomplete information.
- Dominant and dominated strategies; dominant-strategy equilibrium; Pareto optimality.
- Iterated elimination of strictly dominated strategies, applied to spatial competition.

Next lecture

- Best responses and rationalizability.
- Nash equilibrium and mixed strategies.